# Exploring the Replication Files of an Experimental Study (THE I - Beginner)

**Research Design and Methods in Quantitative Research**

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## Instructions

Please read and follow the guidelines below carefully. Then, complete the exercises and report the results in a Quarto document. Compile the Quarto document in PDF and submit both the compiled PDF and `.qmd` files within the deadline.

Further instructions about the submission are below.

### Preparation step 1: Install R and RStudio

To complete this exercise, you will need **R** and **RStudio**. If you don't have them installed already, download and install them from:

- R
- RStudio

A tutorial on how to start using R and R Studio is here. Please **contact the lecturer** and collaborate with your classmates if you have any doubts or need help.

### Preparation step 2: Prepare a Quarto Document

Open RStudio, create a new Quarto document (`.qmd`), and set the output format to PDF. Make sure your Quarto installation is up-to-date:

```
# Install Quarto if needed
## Run this line in a separate script or the Quarto document will not compile
# install.packages("quarto")
```

You can find help on how to set up a Quarto document here.

### Preparation step 3: Read the Assigned Paper and Download the Replication Files

You will need to read the paper *Instrumentally Inclusive: The Political Psychology of Homonationalism*, which you can find in the exercise materials folder.

After reading the paper, download the Study 1 replication materials, also provided in the materials folder:

- Data file: `study1_data.csv`
- R script: `study1.R`

These files have been downloaded from the authors' replication repository. Take a moment to explore it—next time, you may need to download the files yourself!

If you look at their repository, you will notice that the replication materials **do not include a codebook** for the data, so you will need to navigate their R script to identify the relevant variables.

## Exercises

### Exercise 1: Summary of the Paper and Main Findings

Provide a brief summary of the paper and describe its main findings, especially those related to Study 1. Remember to be concise—two or three paragraphs are usually enough for a good summary.

### Exercise 2: Data Exploration I

Load the data file `study1_data.csv` into R and describe the dataset: how many variables and observations does it contain?

Then, describe the experimental variables `treatment` and `imm_1` (i.e., pre-treatment attitudes toward immigration). Convert treatment into a factor (categorical) variable and use a bar plot to visualize its distribution. Next, use a histogram and a kernel density plot to visualize the distribution of immigration attitudes. What do these visualizations tell us about the data?

### Exercise 3: Data Exploration II

Select up to four covariates (e.g., gender, age, etc.) and plot their distributions as well. What do these visualizations tell us about the data?

**Importantly**, you may need to clean (e.g., recode or remove missing values) or transform (e.g., convert a numeric into a categorical variable) some of the variables to produce appropriate visualizations. Document any changes made.

### Exercise 4: Check Covariate Balance

Check for balance across covariates by examining whether the mean values of your selected covariates are similar between the treatment and control groups. Report the results in a table.

***Explain*** your findings. Why would you expect randomization to lead to balance across covariates?

## Submission guidelines

Please submit both the PDF file and the `.qmd` file. Both files should report all the code used for analysis and annotations explaining each step.

The name of the files must follow the structure *take-home_exercise_i_beginner_YOURSURNAME(S).pdf* and *take-home_exercise_i_beginner_YOURSURNAME(S).qmd*, respectively. They should be uploaded to the folder *Students responses/Take-home exercises/Take-home exercise I* in OLAT.

Deadline: **29.10.26**

## References

Turnbull-Dugarte, S. J., & Ortega, A. L. (2024). Instrumentally inclusive: the political psychology of homonationalism. *American Political Science Review, 118(3)*, 1360-1378.